

Thesis for Doctor of Philosophy

Preparation and Mechanical
Properties of TPS(Thermoplastic
Starch)/Poly(Butylene Succinate)
Blends

June 2026

Graduate School

Kumoh National Institute of Technology

Department of Computer·AI Convergence Engineering

Student Name

본 논문작성자는 한국정부초청장학금
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June 2026

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Abstract

This sample abstract demonstrates the required layout for the English abstract. Replace this paragraph with a concise summary of the research purpose, methods, main results, and conclusions. The abstract should be understandable without reading the full thesis and should avoid citations unless they are essential.

TPS(Thermoplastic Starch)와 Poly(Butylene Succinate)의 Blends 제조와 물리적 특성

학생 이름

대학원

국립금오공과대학교

요 약

이 예시 요약은 국문 초록의 형식을 확인하기 위한 자리표시자입니다. 실제 논문에서는 연구 목적, 연구 방법, 주요 결과, 결론을 간결하게 작성하십시오. 국문 요약은 영문 초록과 같은 핵심 내용을 담되, 제출 시점의 대학원 작성 지침에 맞추어 분량과 표현을 조정하십시오.

TABLE OF CONTENTS

List of Tables	viii
List of Figures	ix
List of Notations	x
List of Abbreviations	xi
Chapter I: Introduction and Background	1
1.1 Section	1
1.1.1 Example Subsection	1
1.1.1 A Example Subsubsection	1
1.2 Citation	1
Chapter II: Methods	2
Chapter III: Results	5
Chapter IV: Discussion	8
Chapter V: Conclusion	9
References	9
Appendix A: Experimental Equipment	13
Appendix B: Data Processing	15

LIST OF TABLES

3.1	This is an example portrait table.	6
3.2	This is an example landscape table.	7

LIST OF FIGURES

2.1	Portrait figure	3
2.2	Landscape figure	4
A.1	Stage of a bubble	14

LIST OF NOTATIONS

This page summarizes recurring mathematical notation used throughout the thesis. Add only symbols that appear repeatedly; define local variables in the text where they first appear.

Notation	Meaning
General Notation	
x	Scalar variable.
$\mathbf{x}$	Vector variable.
$\mathbf{X}$	Matrix or tensor variable.
$\mathcal{D}$	Dataset used in the study.
N	Number of samples or observations.
t	Time index or iteration index.
Example Model Notation	
$f_{\theta}(\cdot)$	Model parameterized by θ .
$\hat{y}$	Predicted output.
$\mathcal{L}$	Objective or loss function.
α	Learning rate or weighting coefficient.

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
KIT	Kumoh National Institute of Technology
PDF	Portable Document Format
TPS	Thermoplastic Starch

CHAPTER I

INTRODUCTION AND BACKGROUND

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1.1 Section

This is a section in Chapter 1.

1.1.1 Example Subsection

This is a subsection in Chapter 1.

1.1.1 A Example Subsubsection

This is a subsubsection in Chapter 1.

1.2 Citation

Citations are Kernighan and Ritchie [1], Burdell [2], Einstein et al. [3], Piaget [4], Einstein [5], Knuth [6] and Dirac [7].

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CHAPTER II

METHODS

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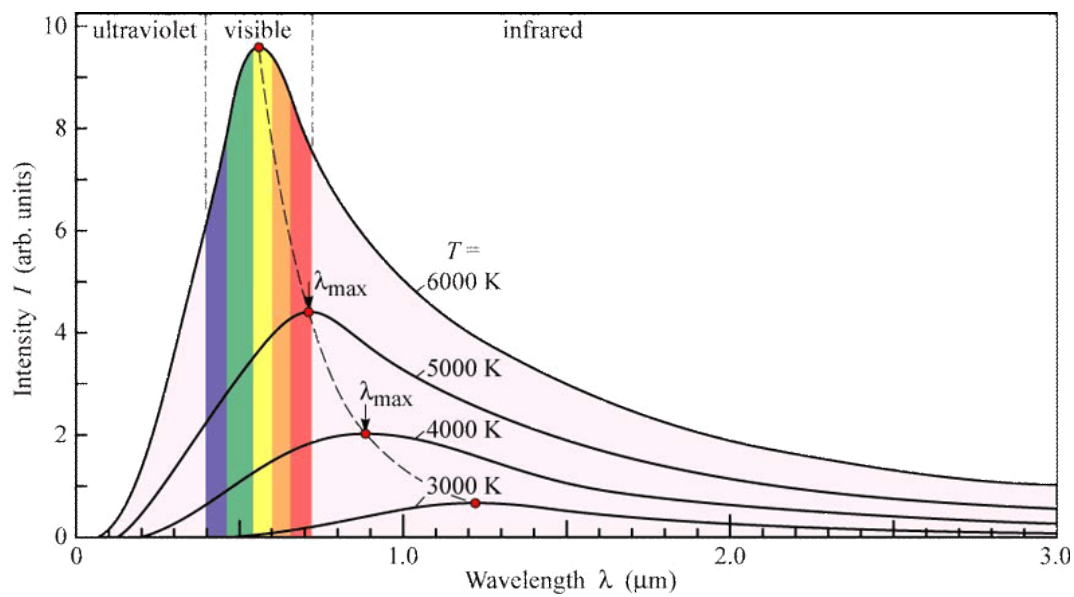


Figure 2.1: Portrait figure

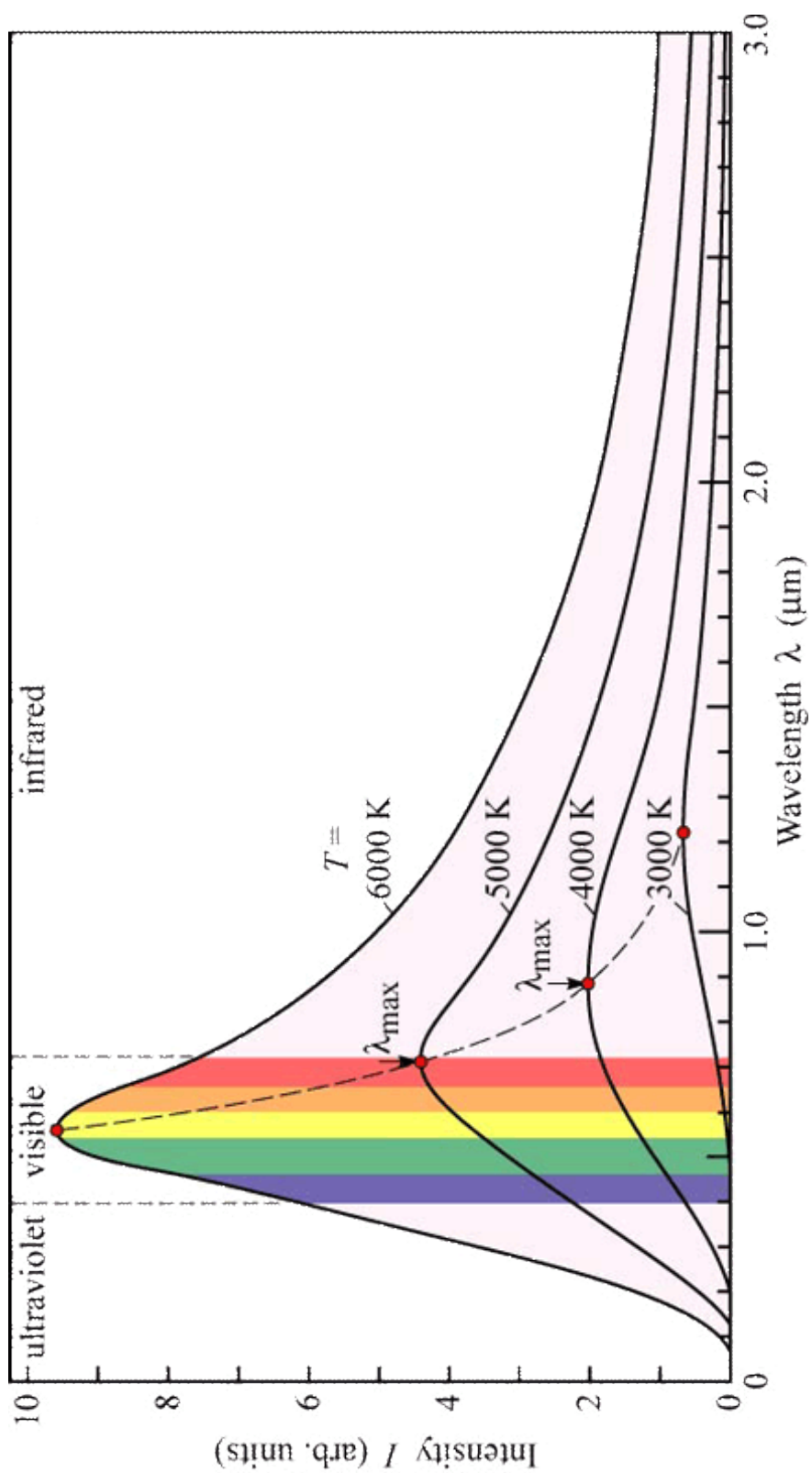


Figure 2.2: Landscape figure

CHAPTER III

RESULTS

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Table can be cited as Table 3.1 or Table 3.2. The portrait table is Table 3.1, and landscape table is Table 3.2.

x	f(x)	g(x)
1	6	4
2	6	3
3	6	2
4	6	2

Table 3.1: This is an example portrait table.

x	f(x)	g(x)	A	B	C	D	E	F	G	H	I	J
2	6	3	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
1	6	4	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
3	6	2	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
4	6	2	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0

Table 3.2: This is an example landscape table.

CHAPTER IV

DISCUSSION

Equation is cited as Equation (4.1) and Equation (4.3) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat Kernighan and Ritchie [1]. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur Burdell [2]. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum Einstein et al. [3].

$$F = ma \tag{4.1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f} \tag{4.2}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{4.3}$$

CHAPTER V

CONCLUSION

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REFERENCES

- [1] Brian W. Kernighan and Dennis M. Ritchie. *The C Programming Language Second Edition*. Prentice-Hall, Inc., 1988.
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ACKNOWLEDGEMENTS

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Appendices

APPENDIX A

EXPERIMENTAL EQUIPMENT

In Figure A.1, Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

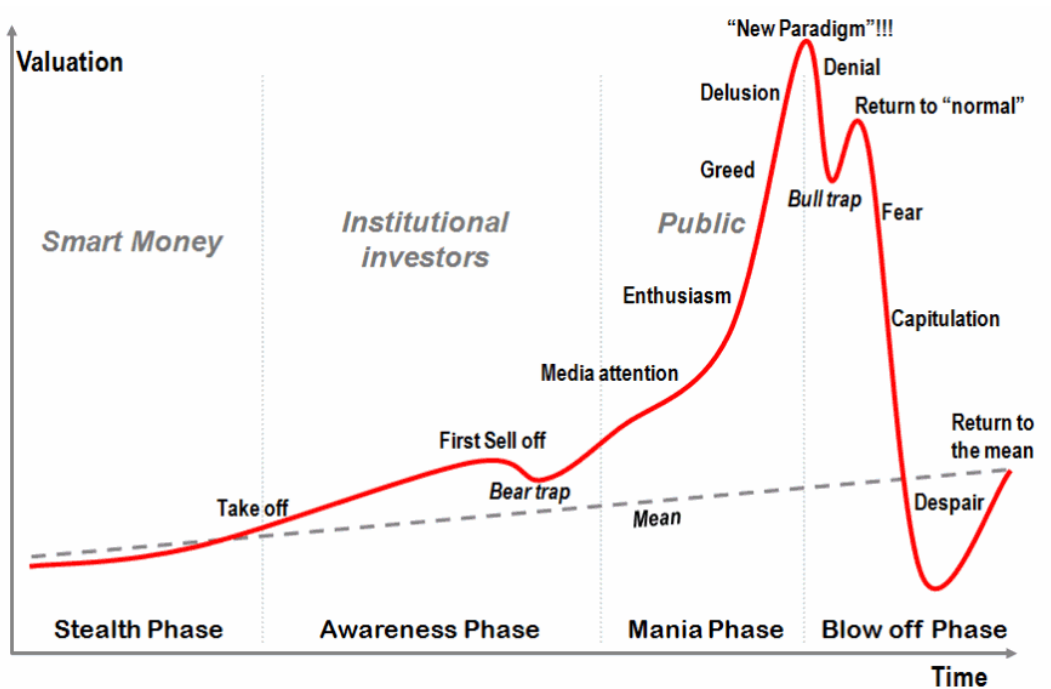


Figure A.1: Stage of a bubble

APPENDIX B

DATA PROCESSING

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Monospace font Test

PPO Algorithm in Pseudocode.

Bold and Monospace Test

PPO Algorithm in Pseudocode.

Algorithm 1 PPO

```
1: for iteration = 1, 2, ... do
2:   for actor = 1, 2, ..., N do
3:     Run policy  $\pi_{\theta_{old}}$  in environment for T time steps
4:     Compute advantage estimates  $\hat{A}_1, \dots, \hat{A}_T$ 
5:   end for
6:   Optimize surrogate L wrt.  $\theta$ , with K epochs and minibatch size  $M \leq NT$ 
7:    $\theta_{old} \leftarrow \theta$ 
8: end for
```
